Mini-Project: Parallel Fractal Generation – Mandelbrot Set

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1 Introduction

The Mandelbrot set1 is the set of values of c in the complex plane for which the orbit of 0 under iteration of the quadratic map $z_{n+1} = z_n^2 + c$ remains bounded. It can be used to produce complex fractal shapes. Since every point is computed independently from the others, it is easy to parallelize the computation of the fractal shapes.

2 Implementation

The code can be found on my GitHub.

As a starting point, I started with the source code of the 4^{th} version of the Applet on http://java.rubikscube.info/: *Applet with Assynchronous Interlaced Double-buffering.*

2.1 Timing

I added a timer and displayed it on the status bar to be able know how long it takes the program to render the image. The timer only measures how long it takes to compute the color for each pixel and excludes the rendering.

2.2 Refactoring

I proceeded with some refactorings that did not change the functionality of the program but rather simplified it such that making the code multi-threaded would become easier. First, I extracted a ColorManager class to clean up the main Mandel class. Then, instead of drawing the pixels block by block, I changed the program to draw the pixels one-by-one from top-left to bottom-right. Next, I extracted a PixelPainter class that is in charge of the painting / coloring of the pixels and that can run as a Runnable in a separate thread.

2.3 Multi-Threading and synchronization

I used an ExecutorService to manage running the PixelPainter threads. I split the work up by rows: each thread roughly paints (num rows / num threads) rows. For synchronization, I used a CountDownLatch: I initialize it to the number of threads and await on it after all threads are launched. In each thread, I call latch.coundDown() once the task is done.

3 Evaluation

The program runs roughly 2x faster with 4 threads vs with 1 thread.

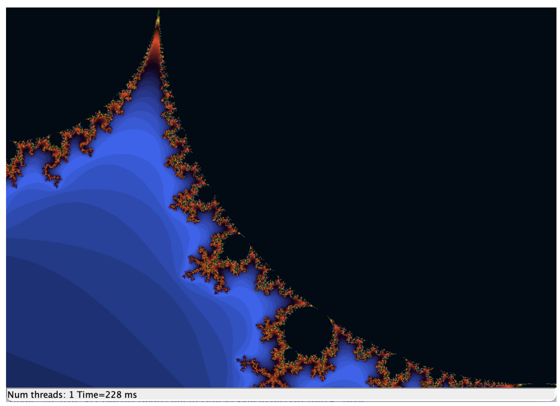

Figure 1: Mandelbrot set rendered with 1 thread: 228ms

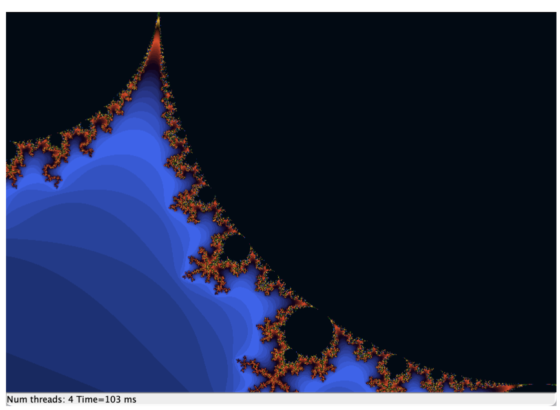

Figure 2: Mandelbrot set rendered with 4 threads: 103ms